UE26CY151A · ENGINEERING CHEMISTRY

SEMESTER 1 · UNIT 1

02

KEY POINTS

Important Points

Definitions, statements of laws, selection rules, “why” reasoning and exam traps, with past-paper questions per topic.

M1
Molecular Spectroscopy
8 topics

M2
Phase Equilibria
5 topics

M3
Computational Chemistry
1 topic

Exam weight	★★★ high · ★★☆ medium · ★☆☆ low
Flags	NOT ON CURRENT SLIDES · TRANSCRIPT ONLY · ADDED FOR CLARITY
Sources	S004·8 slide 8 of deck 004 · T009 class transcript · PDF Unit-1 slide PDF · ESA past paper

Compiled from PESU AV-summary transcripts, class slides and past ESA papers · 13 October 2026

Contents

M1 Molecular Spectroscopy

T1 EM radiation & interaction with matter ★★★

T2 Electromagnetic spectrum & regions ★★★

T3 Born–Oppenheimer approximation ★★☆

T3b Beer–Lambert law ★☆☆

T4 Rotational (microwave) spectroscopy ★★★

T5 Vibrational (IR) – SHO ★★★

T6 Vibrational – anharmonic ★★☆

T7 Electronic spectroscopy & Franck–Condon ★★★

M2 Phase Equilibria

T8 Basic terms & phase rule ★★★

T9 Derivation of phase rule ★★★

T10 One-component system: water ★★★

T11 Two-component: reduced phase rule & Pb–Ag ★★★

T12 Thermal analysis / cooling curves ★★☆

M3 Computational Chemistry

T13 Computational chemistry ★☆☆

MODULE 1

Molecular Spectroscopy

T1 · EM radiation & interaction with matter | T2 · Electromagnetic spectrum & regions | T3 · Born–Oppenheimer approximation | T3b · Beer–Lambert law | T4 · Rotational (microwave) spectroscopy | T5 · Vibrational (IR) – SHO | T6 · Vibrational – anharmonic | T7 · Electronic spectroscopy & Franck–Condon

T1 EM radiation & interaction with matter ★★★

DEFINITION

Molecular spectroscopy: the branch of science dealing with the interaction of electromagnetic radiation with matter (molecules).

★★☆ S001·3

KEY POINT

Spectrophotometer: source → monochromator (entrance slit, dispersive element, exit slit) → sample → detector → recorder.

★★☆ S001·3

KEY POINT

EM radiation has dual character (photons moving as waves); E and B fields oscillate perpendicular to each other and to the direction of propagation; all travel at the speed of light.

★★☆ S001·4

KEY POINT

Atomic spectra: electronic transitions between atomic orbitals → line spectra. Molecular spectra: electronic (HOMO→LUMO) plus vibrational and rotational transitions → band spectra.

★★☆ S001·6-7

KEY POINT

Absorption: molecule goes to a higher level taking $\Delta E = h\nu$; emission: falls to a lower level releasing $h\nu$.

★☆☆ S001

T2 Electromagnetic spectrum & regions ★★★

KEY POINT

Radio (10 m–1 cm): NMR, ESR – nuclear / electron spin reversal. Microwave (1 cm–100 μm): rotational. IR (100 μm–1 μm): vibrational. UV-Vis (1 μm–10 nm): electronic, valence electrons. X-ray (10 nm–100 pm): inner electrons. γ (100–1 pm): nuclear rearrangement.

★★★ S002·4-5

WHY

Rayleigh scattering: scattered light has the same frequency as the incident light (elastic). Raman: shifted frequency — Stokes lines (lower frequency, molecule gains energy), anti-Stokes lines (higher frequency, molecule loses energy).

★★☆ T002 TRANSCRIPT ONLY ADDED FOR CLARITY

ASKED IN PAST ESA PAPERS

(×2) Rayleigh vs Raman; Stokes vs anti-Stokes.

T3 Born–Oppenheimer approximation

★★☆

LAW

> *Born–Oppenheimer approximation: electronic, vibrational and rotational energies of a molecule are independent; $E = E_{rot} + E_{vib} + E_{elec}$ (E_{trans} negligible).*

★★☆ S003·4

KEY POINT

$E_{rot} \ll E_{vib} \ll E_{elec}$ because electronic levels are farther apart than vibrational, which are farther than rotational.

★★☆ S003·4

EXAM TRAP

Mnemonic: Rot < Vib < Elec ↔ Microwave < IR < UV-Vis.

★☆☆ study aid ADDED FOR CLARITY

ASKED IN PAST ESA PAPERS

(×3) State Born–Oppenheimer; energy-level diagram.

T3b Beer–Lambert law

★☆☆

KEY POINT

Beer–Lambert law: absorbance $A = \log(I_0/I) = \varepsilon cl$ — proportional to concentration c and path length l; ε is the molar absorptivity.

★☆☆ ESA Dec-24 NOT ON CURRENT SLIDES ADDED FOR CLARITY

ASKED IN PAST ESA PAPERS

Dec-24 (older syllabus) State the Beer–Lambert law.

T4 Rotational (microwave) spectroscopy

★★★

DEFINITION

Gross selection rule (rotational): molecule must have a permanent dipole moment. Specific rule: $\Delta J = \pm 1$.

★★★ S004·3,10

WHY

HCl and CO are microwave-active (permanent dipole); H_2, Cl_2 and CO_2 are not (no permanent dipole).

★★★ S004·5

KEY POINT

Rigid-rotor model: heteronuclear diatomic = two masses joined by a rigid massless rod; J = 0,1,2...; levels 0, 2B, 6B, 12B...

★★★ S004·6-9

KEY POINT

First line is at 2B (not B) and consecutive lines are 2B apart; B → I → r_0 via $I = \mu r_0^2$.

★★★ S004·12

KEY POINT

Intensity rises then falls with J because higher levels are populated at room temperature (thermal population).

★★☆ S004·12

EXAM TRAP

Traps: J starts at 0; first line = 2B; use μ in kg (not amu); use c in cm s^{-1} when B is in cm^{-1}.

★★★ exam

ASKED IN PAST ESA PAPERS

2023–25 (×4) Derive $I = \mu r_0^2$ for a diatomic rigid rotor; calculate bond length from first line (HCl 21.18 cm^{-1}; HI 13.022 cm^{-1}); draw levels to J=5/6 and the spectrum.

T5 Vibrational (IR) – SHO ★★★

DEFINITION

Gross selection rule (IR): the vibration must change the dipole moment; Δv = ±1.

★★★ S005·3,9

WHY

CO_2 has no permanent dipole (microwave-inactive) but is IR-active: asymmetric stretch and bending change the dipole; symmetric stretch does not.

★★★ S005·4-5

WHY

HI is IR-active but H_2 and I_2 are not (homonuclear: no dipole change).

★★☆ S005·4

WHY

SHO shows only one line: levels are equally spaced and Δv = ±1, so every allowed transition has the same energy $\bar{\nu}_{osc}$.

★★★ S005·9

KEY POINT

Zero-point energy $\varepsilon_0 = \frac{1}{2}\bar{\nu}_{osc} \neq 0$: the molecule never rests even at v = 0. Bond treated as a spring obeying Hooke's law → parabolic PE.

★★★ S005·6-9

KEY POINT

Every bond has a characteristic frequency, so an IR spectrum is a molecular 'fingerprint'.

★☆☆ S005·6

ASKED IN PAST ESA PAPERS

2023–25 Calculate $\bar{\nu}$ and ZPE from k (HF, NO); k from $\bar{\nu}$ (CO); why only one line in SHO?

T6 Vibrational – anharmonic ★★☆

WHY

Why SHO fails: real bonds dissociate when stretched; the true potential is anharmonic (Morse), so levels converge and overtones appear.

★★☆ S006·3

KEY POINT

Anharmonic: $\Delta v = \pm1, \pm2, \pm3...$; most transitions start from $v = 0 \rightarrow$ fundamental ($0 \rightarrow 1$), first overtone ($0 \rightarrow 2$), second overtone ($0 \rightarrow 3$).

★★☆ S006·6

KEY POINT

Hot bands: from $v = 1$ at higher temperature; $1 \rightarrow 2$ appears to the left of the fundamental.

★★☆ S006·9

KEY POINT

x_e is small and positive; line spacing decreases with increasing v.

★★☆ S006·4

ASKED IN PAST ESA PAPERS

2023–25 Find x_e and $\bar{\omega_e}$ from fundamental and overtone; selection rule; failure of SHO.

T7 Electronic spectroscopy & Franck–Condon ★★★

KEY POINT

Electronic transitions: HOMO → LUMO ($\sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$) in the UV-Vis; all molecules show them; visible transitions give colour.

★★★ S007·3

KEY POINT

Vibrational changes give 'coarse structure', rotational changes give 'fine structure'. Homonuclear molecules (H_2, N_2) show vibration/rotation only through electronic spectra.

★★☆ S007·5

KEY POINT

No vibrational selection rule during electronic transitions; most start at $v'' = 0$; lines labelled (v',v'') form a progression converging at high energy (anharmonic excited state).

★★★ S007·6-8

LAW

> *Franck–Condon principle: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition." Hence transitions are vertical.*

★★★ S007·13

KEY POINT

Case (a) $r_e' = r_e''$: (0,0) strongest. Case (b) $r_e' > r_e''$: strongest line (v′,0) with v′>0. Case (c) $r_e' \gg r_e''$: transition goes beyond the dissociation limit → continuum.

★★★ S007·15-16

WHY

A continuum means dissociation: the atoms fly apart with unquantised kinetic energy. The excited state usually has a longer bond length because of antibonding character.

★★★ S007·10,16

ASKED IN PAST ESA PAPERS

2023–25 (×5) State the Franck–Condon principle; draw the three cases with spectra; meaning of a continuum.

MODULE 2

Phase Equilibria

T8 · Basic terms & phase rule | T9 · Derivation of phase rule | T10 · One-component system: water | T11 · Two-component: reduced phase rule & Pb–Ag | T12 · Thermal analysis / cooling curves

T8 Basic terms & phase rule ★★★

DEFINITION Phase (P): homogeneous, physically distinct, mechanically separable part of a system bounded by a surface. Gases: P = 1; miscible liquids: P = 1; each solid / polymorph is a phase.

★★★ S008·4

DEFINITION Component (C): smallest number of independently variable constituents by which the composition of every phase can be expressed (zero/negative values allowed).

★★★ S008·6

DEFINITION Degrees of freedom (F): minimum number of intensive variables (T, P, concentration) to be fixed to define the system completely.

★★★ S008·10

LAW

> *Gibbs phase rule: provided equilibrium is not influenced by external forces (gravity, electrical, magnetic), $F = C - P + 2$.*

★★★ S008·12

KEY POINT Counting: ice/water/vapour P=3, C=1; $CaCO_3 \rightleftharpoons CaO + CO_2$: P=3, C=2; $NH_4Cl \rightleftharpoons NH_3 + HCl$: P=2, C=1 (C=2 if extra HCl/NH_3 added); $Na_2SO_4 \cdot 10H_2O$ system C=2.

★★☆ S008·5-9

KEY POINT Systems minimise Gibbs energy at constant T, P; at equilibrium G is equal in coexisting phases.

★☆☆ S008·3

ASKED IN PAST ESA PAPERS

(×2) Components and phases for $CaCO_3/MgCO_3$ decomposition.

T9 Derivation of phase rule ★★★

KEY POINT At equilibrium all phases have the same T and P and the chemical potential of each component is equal in every phase; mass flows from higher to lower μ.

★★★ S009·3-4

KEY POINT

Derivation: variables = P(C−1) composition terms + 2 (T,P); constraints = C(P−1); F = C−P+2.

★★★ S009·7

ASKED IN PAST ESA PAPERS

2023–25 (×3) Derive the phase rule for C components and P phases.

T10 One-component system: water ★★★

KEY POINT

Water: F = 3−P → one phase F=2 (bivariant area), two phases F=1 (univariant line), three phases F=0 (invariant point).

★★★ S009·8-10

KEY POINT

Triple point O: 0.0098 °C, 4.58 mmHg (F=0). Critical point A: 374 °C, 218 atm; above it water cannot be liquid.

★★★ S009·15

WHY

Fusion curve OC has negative slope: ice melts with decrease in volume (ΔV<0 → dP/dT<0), and it is steep because |ΔV| is small. Sublimation (OB) and vaporisation (OA) slopes are positive and less steep because ΔV is large.

★★★ S009·12-14; T009b TRANSCRIPT-ASSISTED

KEY POINT

Metastable OA′: supercooled water; its vapour pressure is higher than that of ice at the same T; disturbance returns the system to stable OB.

★★★ S009·16

ASKED IN PAST ESA PAPERS

2023–25 (×4) Draw and label the water phase diagram; F in each region; metastable curve; negative slope of melting curve.

T11 Two-component: reduced phase rule & Pb–Ag ★★★

WHY

General phase rule not used for 2 components: C=2 gives F=4−P, F=3 for P=1, needing 3-D; so pressure is fixed (≈1 atm) → F = C−P+1.

★★★ S010·2-3

DEFINITION

Condensed system: gas phase absent, hardly affected by small pressure changes.

★★☆ S010·3

KEY POINT

Pb–Ag: components miscible in all proportions in melt, immiscible as solids; plot is temperature vs composition at constant pressure.

★★★ S010·4

KEY POINT

Freezing-point curves: C=2, P=2 → F=1 (univariant). Eutectic point O: P=3, F=0 (invariant), lowest freezing point, definite composition and sharp melting point.

★★★ S010·7-8

KEY POINT

Numerical facts: Pb mp 327 °C, Ag mp 961 °C, eutectic 303 °C at 97.4 % Pb / 2.6 % Ag.

★★★ S010; PDF

KEY POINT

Desilverisation of argentiferous lead (Pattinson-type process) exploits this: Pb crystallises first, leaving Ag-rich melt.

★☆☆ T010 TRANSCRIPT ONLY

ASKED IN PAST ESA PAPERS

2023–25 Draw the Pb–Ag diagram with eutectic values; why F = C–P+1.

T12 Thermal analysis / cooling curves ★★☆

DEFINITION

Thermal analysis: study of cooling curves (temperature vs time) of various compositions during solidification; freezing and eutectic points are read from them.

★★☆ S012·2

KEY POINT

Pure substance: smooth fall, then a flat halt at the freezing point. Mixture: a change of slope when crystallisation starts, then a halt at the eutectic temperature. Plotting break/halt temperatures against composition builds the phase diagram.

★★☆ S012·2-3

MODULE 3

Computational Chemistry

T13 · Computational chemistry

T13 Computational chemistry ★☆☆

DEFINITION

Computational chemistry: techniques to solve chemical problems by simulating chemical systems, giving atomic-level information, using theoretical chemistry in efficient programs.

★☆☆ S011·3

KEY POINT

First theoretical work: Heitler and London, 1927. (Nobel 1998 Kohn & Pople; 2013 Karplus, Levitt & Warshel.)

★☆☆ S011·3; brief ADDED FOR CLARITY

KEY POINT

Scope: molecular properties (HOMO/LUMO, UV/IR/NMR spectra), reaction mechanism and kinetics, material design, drug discovery.

★☆☆ S011·6

KEY POINT

Classical: molecular mechanics (balls and springs, very fast) and molecular dynamics (motion of atoms). Quantum: semi-empirical, ab initio (wavefunction) and DFT (electron density; faster than ab initio, slower than semi-empirical).

★☆☆ S011·7-12

KEY POINT

Advantages: easier, cheaper, safer than experiments; complements, not replaces, experiment.

★☆☆ S011·13

KEY POINT

Ethylene: sp^2 C; σ + π bond; HOMO = π, LUMO = π*. Steps: geometry → basis set → method → solve → visualise.

★☆☆ S011·14-17

KEY POINT

AI-enhanced: fast screening, property prediction, reaction-pathway mapping, structure generation; speed-up from hours/days to milliseconds after training.

★☆☆ S011·19-20

ASKED IN PAST ESA PAPERS

(×1) Define computational chemistry; three applications.